

PROJECT REPORT

AIR QUALITY INDEX (AQI)

PREDICTION SYSTEM

SKARDU, PAKISTAN

SUBMITTED BY
Shehnaz Haider

June 2026

Table of Contents

Contents

Table of Contents	2
1. Introduction	4
2. Problem Statement	4
3. Data Collection	4
3.1 OpenWeather Air Pollution API	4
3.2 Open-Meteo Historical Weather API	5
3.3 Backfill Strategy	5
4. Exploratory Data Analysis (EDA)	5
4.1 Dataset Overview	5
4.2 Missing Values Analysis	6
4.3 Statistical Summary & Distribution	7
4.4 Outlier Detection	7
4.5 Feature Correlation Analysis	8
4.6 Seasonal & Temporal Patterns	8
5. Data Preprocessing	9
Step 1: Duplicate Removal	9
Step 2: Log Transformation	10
Step 3: Outlier Handling — Winsorization	10
Step 4: Feature Scaling	10
Step 5: Target Variable Creation	10
6. Feature Engineering	10
6.1 Temporal Features	10
6.2 EPA AQI Computation	10
6.3 AQI Change Rate	11
6.4 Lag & Rolling Statistics Features	11
7. Model Training & Evaluation	11
7.1 Linear Regression	11
7.2 Random Forest Regressor	12
7.3 XGBoost	12
7.4 Gradient Boosting	12
7.5 Prophet (Time Series)	12

8. Project Workflow	13
8.1 Data Collection (Hourly).....	13
8.2 Daily Preprocessing & Feature Engineering	13
8.3 Model Training & Prediction (Daily).....	13
8.4 Result Serving via Streamlit Dashboard	13
9. Conclusion and Model Performance	13
References	14

1. Introduction

Air pollution is one of the most pressing environmental challenges of our time, directly impacting human health, climate, and ecosystems. Predicting the Air Quality Index (AQI) helps governments, environmental agencies, and citizens take precautionary measures before air quality reaches hazardous levels.

The aim of this project was to develop a fully automated, end-to-end machine learning-based AQI prediction system using historical air quality and weather data, stored and managed on Hopsworks Feature Store. By training multiple models and comparing their performance, we selected the best-performing model to power a real-time Streamlit dashboard.

2. Problem Statement

The challenge was to predict the AQI for the next three days for Skardu, Pakistan, based on historical AQI and related features including pollutant concentrations and weather measurements.

The system needed to:

- Automatically collect hourly air quality and weather data from external APIs
- Store and manage data in Hopsworks Feature Store
- Preprocess and engineer features for AQI prediction
- Train multiple ML models and compare their performance
- Output EPA-standard AQI predictions for the next 3 days.
- Serve predictions through an interactive Streamlit web dashboard
- Automate the entire pipeline using GitHub Actions CI/CD

3. Data Collection

Data for this project was collected using two complementary APIs to ensure complete and accurate feature coverage:

3.1 OpenWeather Air Pollution API

The primary data source for pollutant concentrations. A GitHub Actions workflow runs every hour, sending requests to the OpenWeatherMap API and retrieving atmospheric measurements for Skardu. The key pollutants collected include:

- SO₂ (Sulfur Dioxide) — emitted from combustion of fossil fuels and biomass
- CO (Carbon Monoxide) — produced by incomplete combustion; main driver of winter AQI in Skardu
- O₃ (Ozone) — secondary pollutant with strong respiratory health impacts
- NO₂ (Nitrogen Dioxide) — traffic and industry-related pollutant

- PM_{2.5} (Particulate Matter $\leq 2.5\mu\text{m}$) — fine inhalable particles; strongest AQI predictor ($r=0.92$)
- PM₁₀ (Particulate Matter $\leq 10\mu\text{m}$) — larger inhalable particles ($r=0.91$ with AQI)
- NH₃ (Ammonia) — contributes to secondary particulate matter formation

3.2 Open-Meteo Historical Weather API

Since OpenWeather does not provide free historical weather data, Open-Meteo (a free, no-API-key service) was used to backfill historical weather measurements. Weather features collected include:

- Temperature (°C) — strong negative correlation with AQI ($r = -0.59$); lower temperatures trap pollutants in Skardu valley
- Humidity (%) — mild negative correlation with AQI ($r = -0.24$)
- Wind Speed (m/s) — weak correlation ($r = -0.03$); valley geography limits wind dispersion

3.3 Backfill Strategy

A historical backfill was performed to collect 6 months of hourly data (November 2025 — May 2026), resulting in 4,135 rows. This was achieved by running the feature pipeline script over the historical date range using OpenWeather's historical air pollution endpoint and Open-Meteo's archive API.

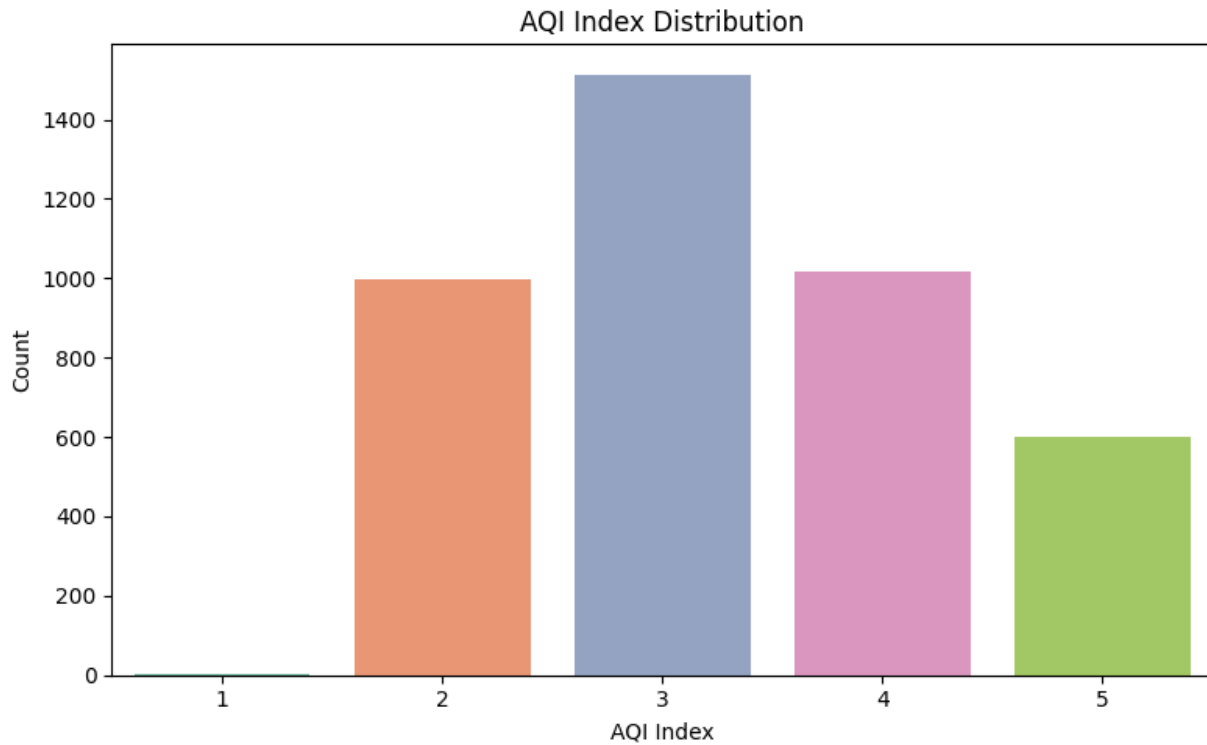
All collected data is stored in Hopsworks Feature Store under the feature group `aqi_predictionv2` (version 1).

4. Exploratory Data Analysis (EDA)

To understand the structure, distribution, and relationships within the dataset, a comprehensive EDA was performed. This step shaped feature engineering and preprocessing decisions.

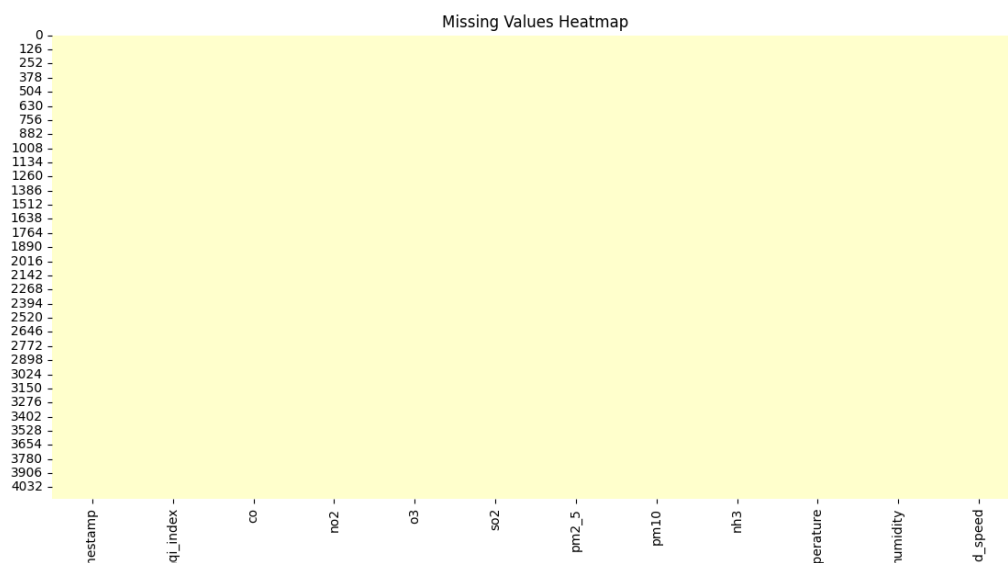
4.1 Dataset Overview

The dataset contained 4,135 hourly air quality readings spanning November 2025 to May 2026. Features included pollutant concentrations (PM_{2.5}, PM₁₀, CO, NO₂, SO₂, O₃, NH₃), weather variables (temperature, humidity, wind speed), and an AQI index (OpenWeather scale 1–5).



4.2 Missing Values Analysis

A missing values heatmap confirmed zero missing values across all 12 features after Open-Meteo weather backfill. The initial backfill using OpenWeatherMap had approximately 4,100 rows with zero temperature, humidity, and wind speed values, which were successfully replaced with Open-Meteo data.



4.3 Statistical Summary & Distribution

Key statistics revealed important characteristics:

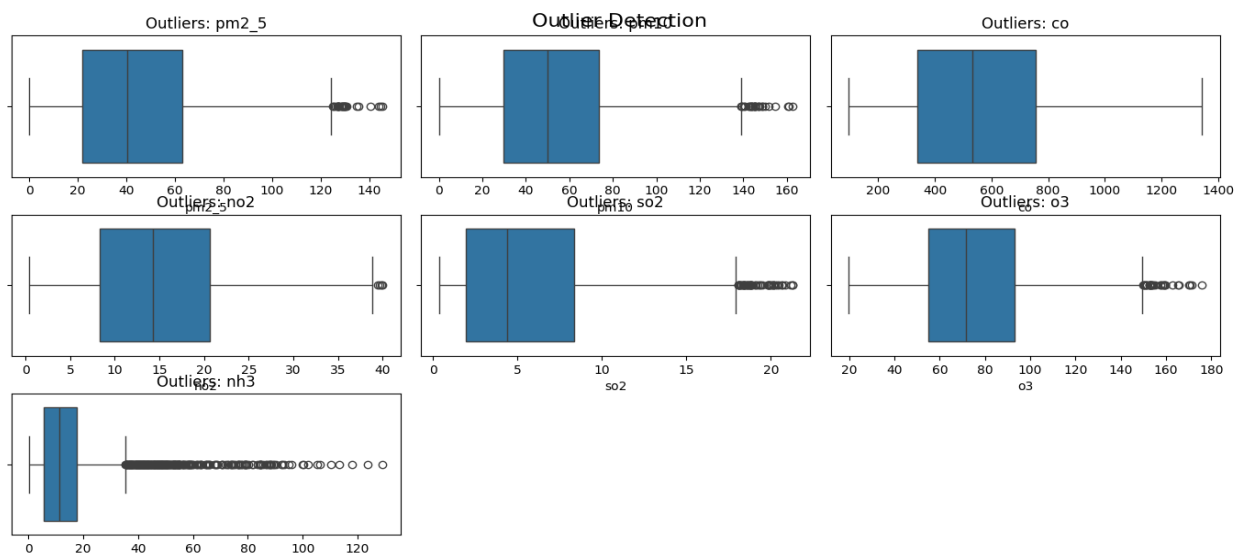
Feature	Min	Mean	Max	Std Dev	Skewness
CO ($\mu\text{g}/\text{m}^3$)	94.73	557.56	1342.93	270.84	High
PM2.5 ($\mu\text{g}/\text{m}^3$)	0.00	44.78	145.57	27.30	Moderate
PM10 ($\mu\text{g}/\text{m}^3$)	0.00	53.80	162.57	29.71	Moderate
NH3 ($\mu\text{g}/\text{m}^3$)	0.35	14.57	129.18	14.35	Very High
Temperature ($^{\circ}\text{C}$)	-15.10	2.76	22.50	6.01	Low
AQI Index (1-5)	1	3.29	5	0.99	Low

Distribution analysis confirmed that CO, NH3, and SO2 exhibit highly right-skewed distributions, requiring log transformation prior to model training.

4.4 Outlier Detection

Boxplot analysis revealed notable outliers in several pollutants:

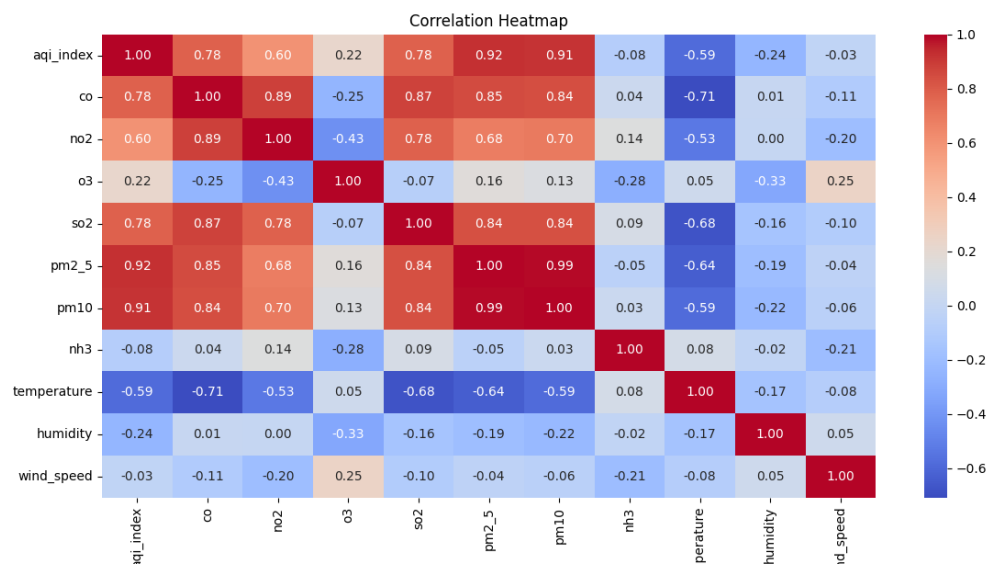
- NH3: Most extreme outliers extending to $129 \mu\text{g}/\text{m}^3$ when median is $\sim 11 \mu\text{g}/\text{m}^3$
- CO: Wide spread from 94 to $1,342 \mu\text{g}/\text{m}^3$ with frequent high-value spikes
- SO2, O3, PM2.5, PM10: Moderate outliers manageable through winsorization



4.5 Feature Correlation Analysis

The correlation heatmap revealed critical relationships that directly informed feature selection:

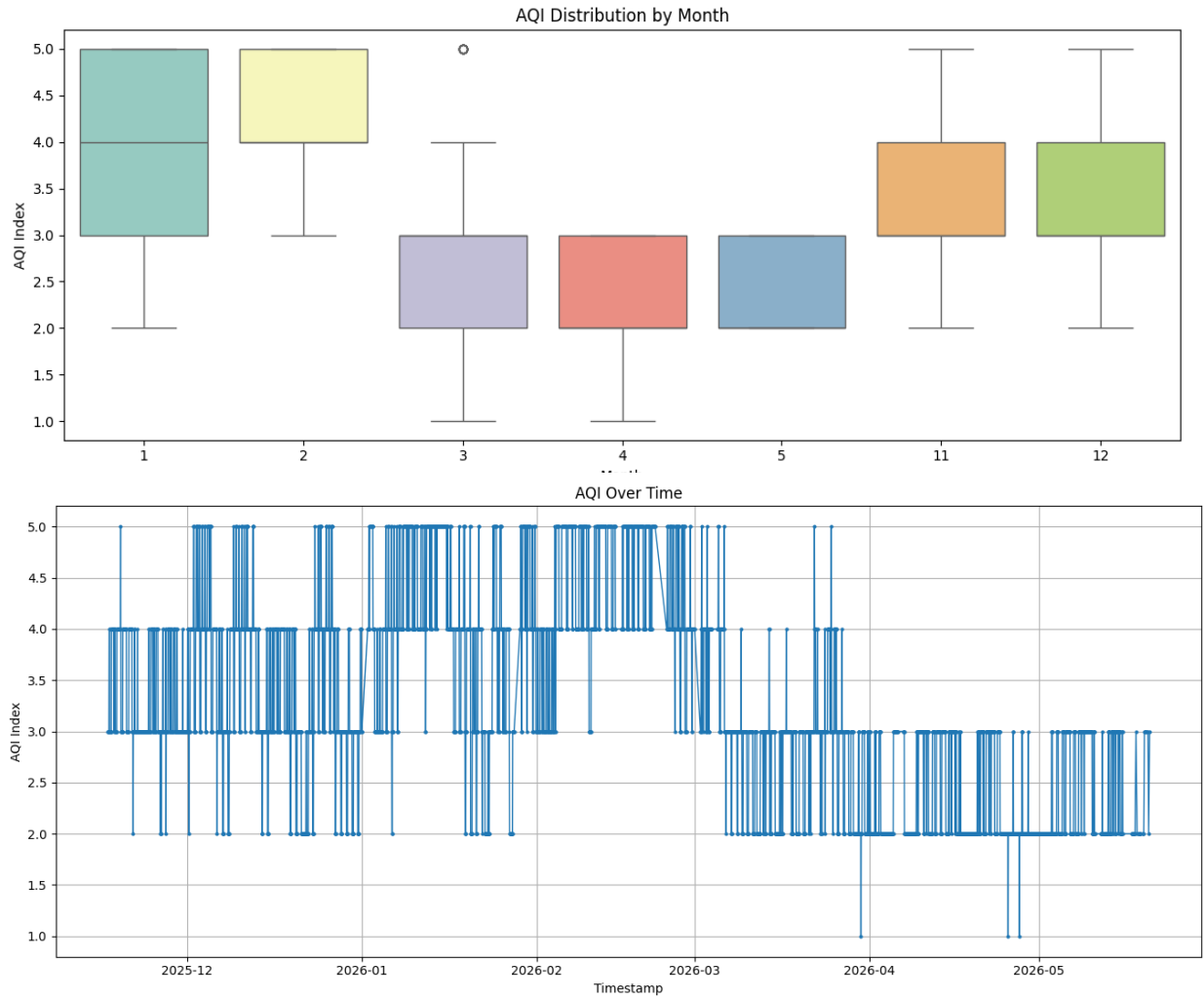
- PM2.5 ($r = 0.92$) and PM10 ($r = 0.91$) — strongest predictors of AQI
- CO ($r = 0.78$) and SO2 ($r = 0.78$) — highly correlated with AQI
- Temperature ($r = -0.59$) — strong negative correlation confirming seasonal effect
- NO2 ($r = 0.60$) — moderate positive correlation
- Humidity ($r = -0.24$) — mild negative correlation
- Wind speed ($r = -0.03$) — negligible correlation due to valley geography
- NH3 ($r = -0.08$) — weak negative correlation; least predictive pollutant



4.6 Seasonal & Temporal Patterns

Strong seasonal patterns were identified from the AQI time series and monthly distribution charts:

- Winter (November–February): AQI predominantly 4–5 (Poor to Very Poor). Cold air inversions trap CO and particulates in Skardu valley.
- Spring (March–May): AQI drops to 2–3 (Fair to Moderate). Warmer temperatures and increased wind disperse pollutants.
- Daily patterns: Hours 7–11 AM show lower AQI; hours 15–18 show peak AQI, consistent with afternoon valley heating effects.



5. Data Preprocessing

Preprocessing decisions were directly informed by EDA findings to ensure the dataset was clean, consistent, and ready for machine learning.

Step 1: Duplicate Removal

Exact duplicate rows (excluding timestamp) were removed to maintain data integrity and prevent repeated observations from biasing model training.

Step 2: Log Transformation

Log transformations (log1p) were applied to CO, SO₂, and NH₃ due to their highly right-skewed distributions identified in EDA. This brings their distributions closer to normality, improving model stability for algorithms sensitive to non-normal distributions.

Step 3: Outlier Handling — Winsorization

A winsorization approach was applied to cap pollutant values at the 1st and 99th percentiles for PM_{2.5}, PM₁₀, O₃, SO₂ (log), NH₃ (log), and CO (log). Winsorization was chosen over outlier removal because it preserves all observations — including real pollution spikes — while minimizing the distortion extreme values cause during training.

Step 4: Feature Scaling

MinMaxScaler was applied to all numerical features, scaling values to the range [0, 1]. This ensures all inputs have comparable magnitudes and prevents high-value features like CO (up to 1,342) from dominating lower-value features like SO₂ (up to 21).

Step 5: Target Variable Creation

Three target columns were created by shifting the calculated EPA AQI forward by 24, 48, and 72 hours respectively:

- aqi_day1: EPA AQI 24 hours ahead (Day 1 forecast)
- aqi_day2: EPA AQI 48 hours ahead (Day 2 forecast)
- aqi_day3: EPA AQI 72 hours ahead (Day 3 forecast)

Rows where future target values do not exist (last 72 rows) were dropped, leaving 4,063 usable training samples.

6. Feature Engineering

6.1 Temporal Features

Three temporal components were extracted from the timestamp:

- Hour (0–23): Captures diurnal pollution patterns — rush hour spikes and overnight temperature inversions
- Day (1–31): Identifies short-term variations and weather event impacts
- Month (1–12): Captures seasonal patterns — the most important temporal feature given Skardu's strong seasonal AQI cycle

6.2 EPA AQI Computation

Rather than using OpenWeatherMap's simplified 1–5 AQI scale, a proper EPA AQI (0–500) was calculated from raw pollutant concentrations using US EPA standard breakpoints. This provides meaningful, internationally recognized AQI values for prediction and display.

The formula used:

$$\text{AQI} = ((\text{AQI_high} - \text{AQI_low}) / (\text{Conc_high} - \text{Conc_low})) \times (\text{Conc} - \text{Conc_low}) + \text{AQI_low}$$

The final AQI for each timestamp is the maximum AQI value across all calculated pollutant AQIs (PM2.5, PM10, CO, NO2, SO2, O3).

6.3 AQI Change Rate

A first-order difference of calculated_aqi was computed as aqi_change_rate. This feature captures short-term fluctuations and allows models to learn from sudden spikes or drops in air quality — particularly useful for detecting the rapid changes associated with weather fronts arriving in Skardu valley.

6.4 Lag & Rolling Statistics Features

To capture temporal autocorrelation — the key insight that today's AQI strongly predicts tomorrow's AQI — the following features were engineered:

Lag features (AQI and PM2.5 values from 1h, 3h, and 24h ago):

- aqi_lag_1h, aqi_lag_3h, aqi_lag_24h
- pm25_lag_1h, pm25_lag_3h, pm25_lag_24h

Rolling statistics (mean and standard deviation over 6h, 12h, and 24h windows):

- aqi_rolling_mean_6h/12h/24h — smoothed AQI trend
- aqi_rolling_std_6h/12h/24h — AQI variability
- pm25_rolling_mean_6h/12h/24h — smoothed PM2.5 trend

These features were the primary reason for R^2 improvement from 0.30 (without lags) to 0.42 (with lags), as they provide the model with crucial context about recent air quality trajectory.

7. Model Training & Evaluation

Five models were trained and evaluated on the processed dataset. The data was split 80/20 (train/test) using a chronological split to preserve temporal order and prevent data leakage. Each model predicts three targets simultaneously: aqi_day1, aqi_day2, and aqi_day3.

All models were evaluated using three metrics averaged across all three forecast horizons:

- RMSE (Root Mean Squared Error): Penalizes large prediction errors
- MAE (Mean Absolute Error): Average absolute prediction error in AQI units
- R^2 (Coefficient of Determination): Proportion of variance explained (1.0 = perfect)

7.1 Linear Regression

A standard multi-output linear regression model was trained as both a baseline and candidate model. Linear regression assumes a linear relationship between features and targets.

Performance: RMSE = 40.20, MAE = 33.79, R^2 = 0.42 — Best overall model.

The fact that linear regression outperformed all complex models is a meaningful finding. It indicates that the relationship between lag features and future AQI is predominantly linear — i.e., if AQI was high in the past 24 hours, it will likely remain high. Complex models overfit noise in the data rather than learning this simple pattern.

7.2 Random Forest Regressor

An ensemble model using multiple decision trees trained with bagging. Handles non-linear relationships and feature interactions well.

Performance: RMSE = 41.61, MAE = 37.86, $R^2 = 0.37$.

Performed slightly worse than linear regression, likely due to overfitting on the relatively small 4,063-row dataset. Day 1 predictions ($R^2 = 0.44$) were notably better than Day 3 predictions.

7.3 XGBoost

Gradient boosted trees using the XGBoost library. Known for strong performance on tabular data with feature interactions.

Performance: RMSE = 41.88, MAE = 35.01, $R^2 = 0.37$.

Competitive performance but did not outperform linear regression. Day 1 R^2 of 0.44 indicates strong short-term prediction capability.

7.4 Gradient Boosting

Scikit-learn's Gradient Boosting implementation, similar to XGBoost but with different regularization characteristics.

Performance: RMSE = 43.61, MAE = 34.43, $R^2 = 0.31$.

Fourth place overall. The MAE of 34.43 is competitive but RMSE suffers from occasional large errors on extreme AQI days.

7.5 Prophet (Time Series)

Prophet was unexpectedly the worst-performing model, likely due to the limited dataset size. Since Prophet relies solely on timestamp patterns and cannot leverage the strong pollutant features (PM2.5, CO) that drive AQI, it failed to produce meaningful predictions and was excluded from the final model comparison.

All trained models were saved to the Hopsworks Model Registry with their evaluation metrics for version tracking and reproducible deployment.

8. Project Workflow

The overall workflow follows a fully automated serverless pipeline using OpenWeather API, Open-Meteo API, Hopsworks Feature Store & Model Registry, GitHub Actions for CI/CD, and Streamlit Cloud for deployment.

8.1 Data Collection (Hourly)

A GitHub Actions workflow runs automatically every hour using a cron schedule. It fetches current pollutant and weather data from OpenWeather, then inserts a new row into the Hopsworks feature group `aqi_predictionv2`.

8.2 Daily Preprocessing & Feature Engineering

A separate GitHub Actions workflow runs daily at 3 AM UTC. It fetches the collected raw data from Hopsworks, applies all preprocessing steps and stores the processed dataset in the Hopsworks feature group `processed_aqi_skardu_v2`.

8.3 Model Training & Prediction (Daily)

Immediately after preprocessing, the training workflow executes five model training scripts sequentially (Linear Regression, Random Forest, XGBoost, Gradient Boosting, Prophet). Each script loads the processed data, trains the model, evaluates performance, and saves the trained model to the Hopsworks Model Registry with performance metrics attached.

8.4 Result Serving via Streamlit Dashboard

The Streamlit web app deployed on Streamlit Cloud connects to Hopsworks, loads the best registered model and the latest processed features, and generates 3-day AQI predictions. The dashboard displays:

- 3-day AQI forecast
- Health advisory alerts for hazardous predictions
- Current weather conditions
- Pollutant-level breakdown with safe-limit indicators

9. Conclusion and Model Performance

The AQI Prediction System successfully automated the end-to-end pipeline for data collection, preprocessing, feature engineering, model training, and prediction serving. By leveraging Hopsworks Feature Store for data management and FastAPI for result delivery, the project demonstrated how real-world environmental monitoring can be efficiently operationalized.

Overall, the project proved that machine learning models, when combined with a structured data pipeline, can provide timely and accurate air quality forecasts. This predictive capability can support policymakers, environmental agencies, and citizens in making informed decision.

References

1. US EPA. Technical Assistance Document for the Reporting of Daily Air Quality — the Air Quality Index (AQI). EPA-454/B-24-002, 2024.
2. Wikipedia. Air quality index. https://en.wikipedia.org/wiki/Air_quality_index
3. OpenWeatherMap. Air Pollution API Documentation. <https://openweathermap.org/api/air-pollution>
4. Open-Meteo. Free Weather API Documentation. <https://open-meteo.com/en/docs/historical-weather-api>